Vorlesung
Informationsvisualisierung

Kapitel 1
EINFÜHRUNG

Lecture
Information Visualization

Chapter 1
INTRODUCTION

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Universität Trier – Sommersemester 2026

Experiment

Two bars can convey a lot of information.

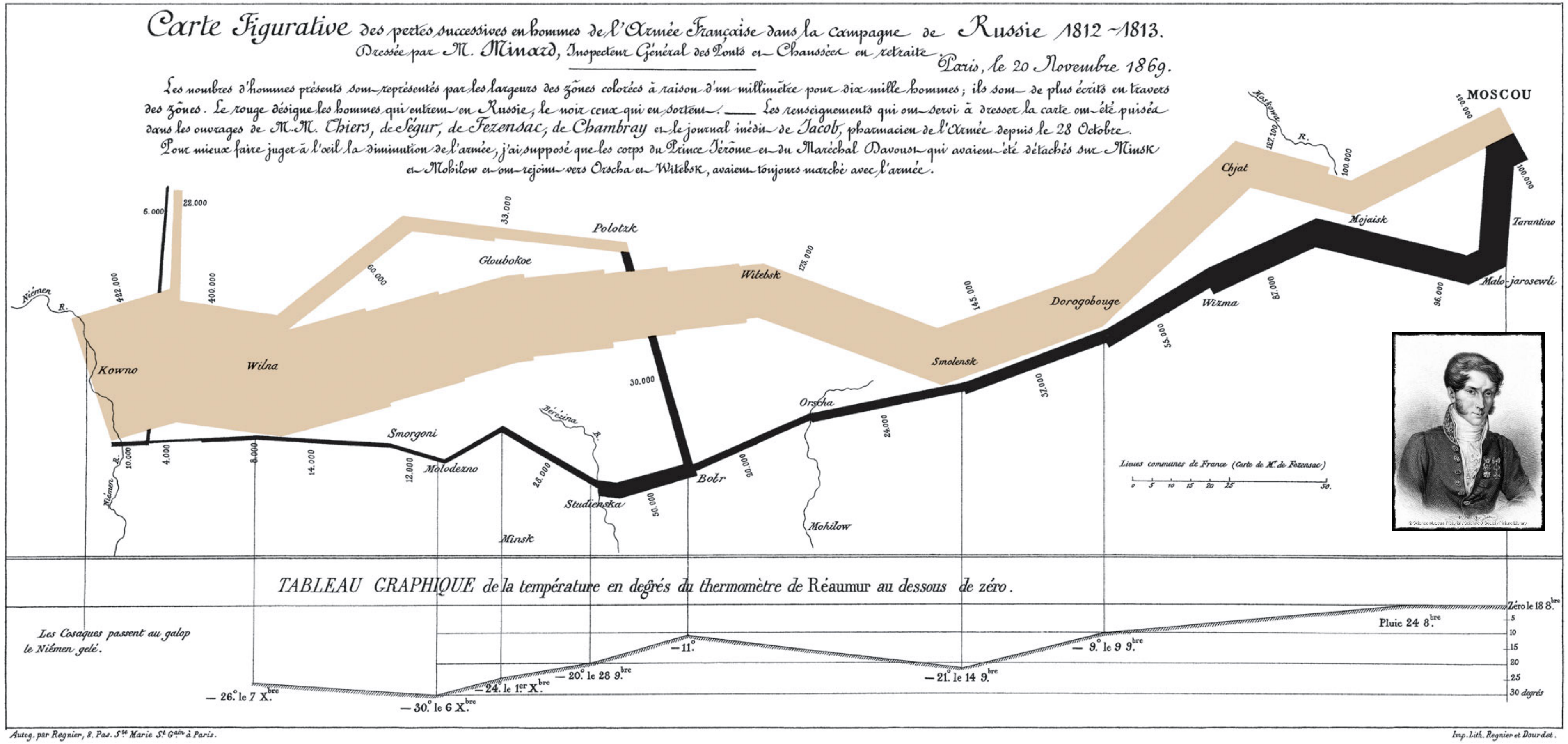
Information Visualization

Some examples …

A historical Example

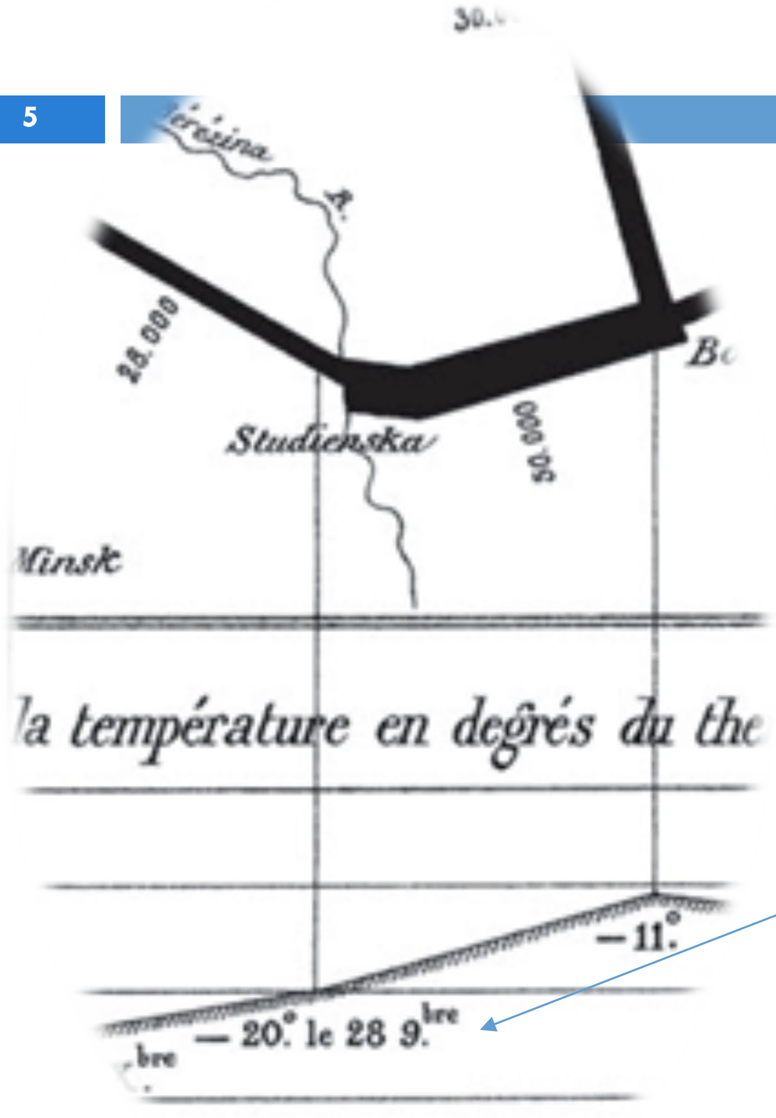
1mm = 6.000 persons

−30 °Réaumur = −37,5 °Celsius



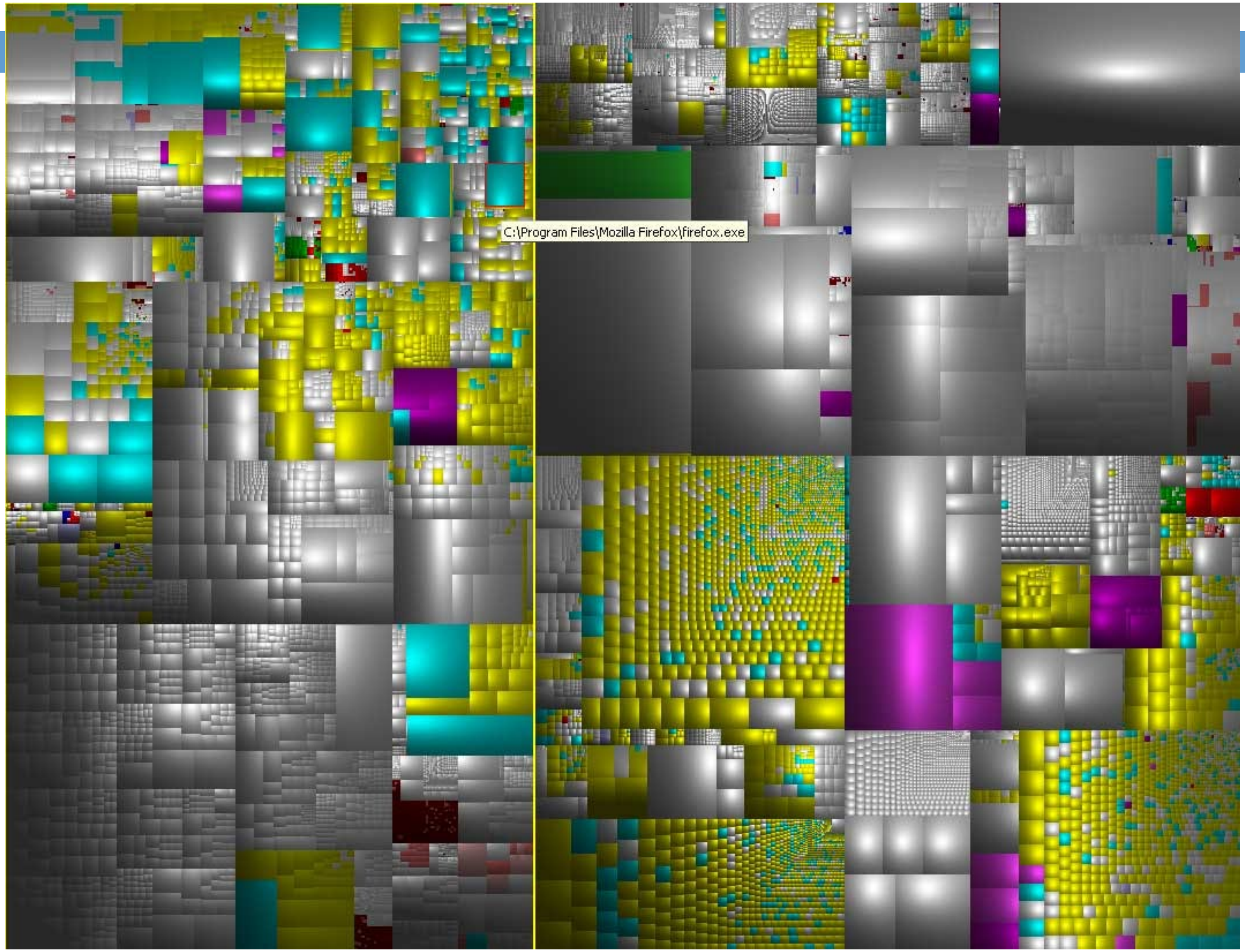
Napoleon's Russian Campaign (Russlandfeldzug)1812-1813,
illustrated by Charles Joseph Minard, 1869

A historical Example



- Line indicates „flow“
- Point on line = geographic location
- Width of line = number of troops
- Time goes from left to right and back again
- Temperature and date

A modern Example

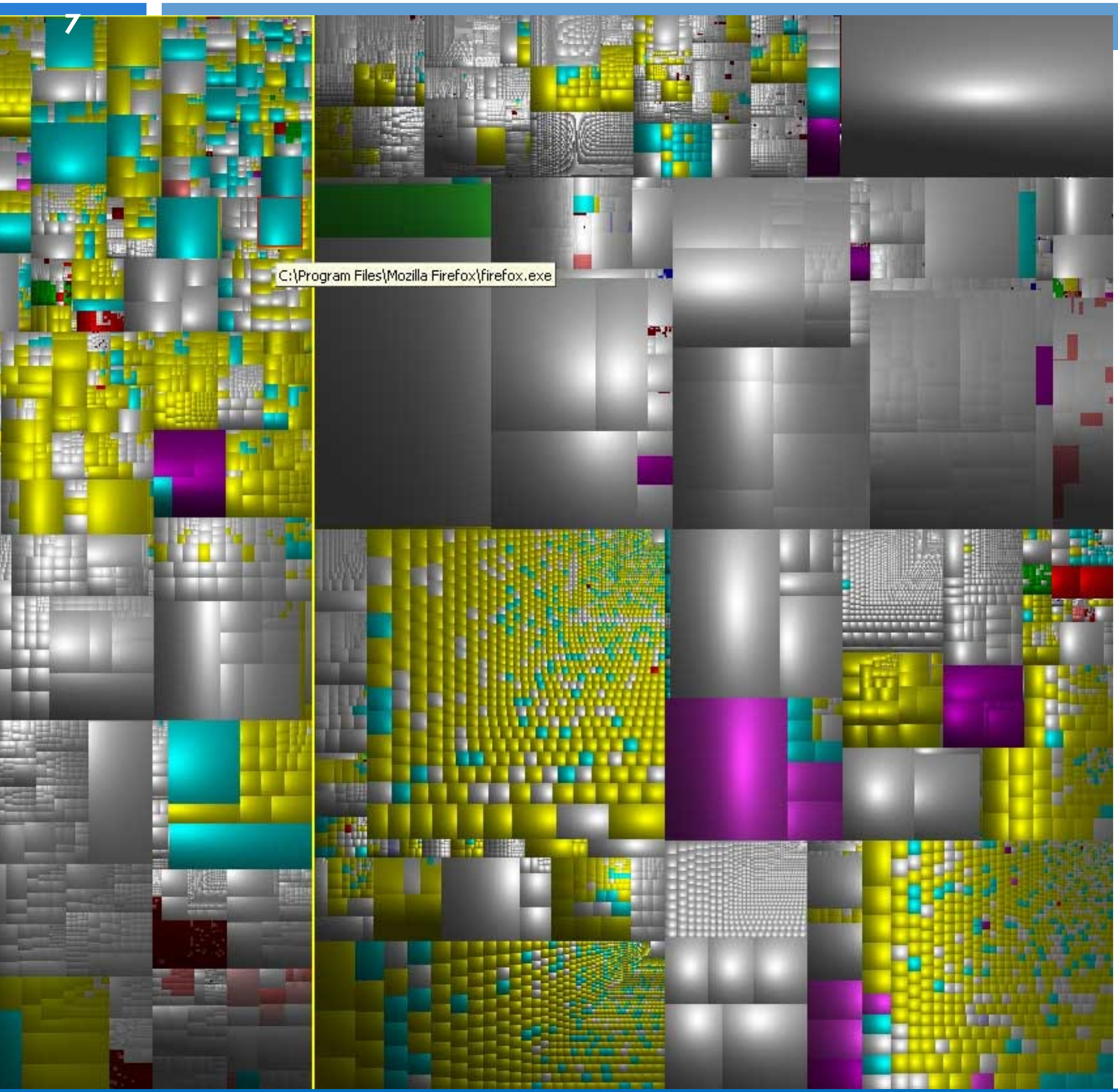


Treemap

(Shneiderman, 1992)

File system visualized as a *Cushion Treemap* in the tool SequoiaView

A modern Example



Treemap (Shneiderman, 1992)

Definition: Information Visulization

Information visualization *is the communication of* ***abstract data*** *through the use of* ***interactive visual interfaces.***

[Keim et al., 2006]

Original quote: “We define information visualization more generally as the communication of abstract data relevant in terms of action through the use of interactive visual interfaces.“

Brainstorming

Examples of

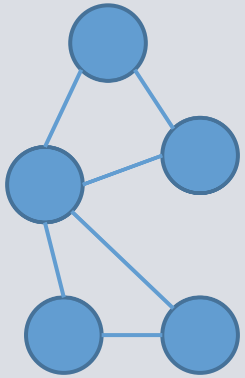
„abstract data“?

Abstract Data

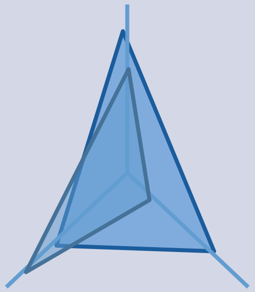
Text and Tables

Information
Tag Text
Visualisation
Table Cloud
Document abstract
Meta-Data

Hierarchies and Graphs

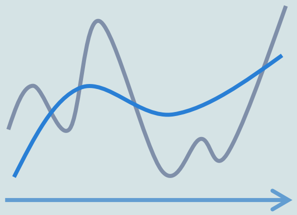


Multivariate Data

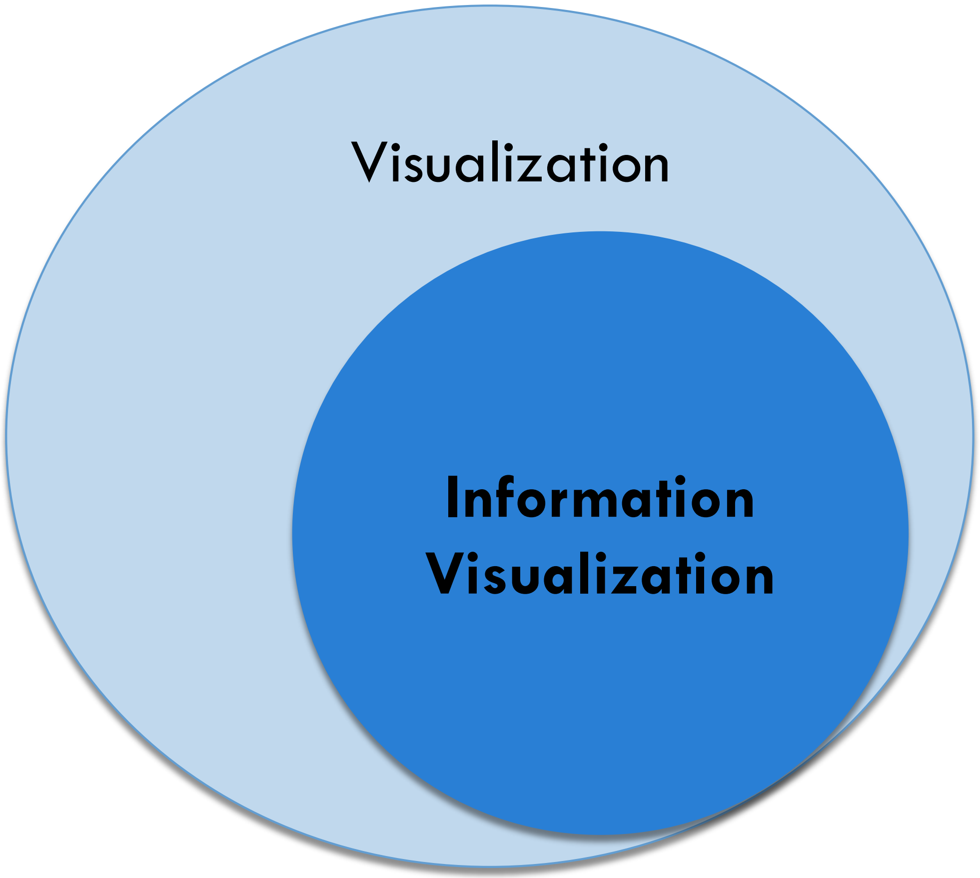


n-dimensional

Time Series (Zeitreihen)



Visualization vs. Information Visualization



Visualization =

the use of computers or techniques for comprehending data or to extract knowledge from the results of simulations, computations, or measurements

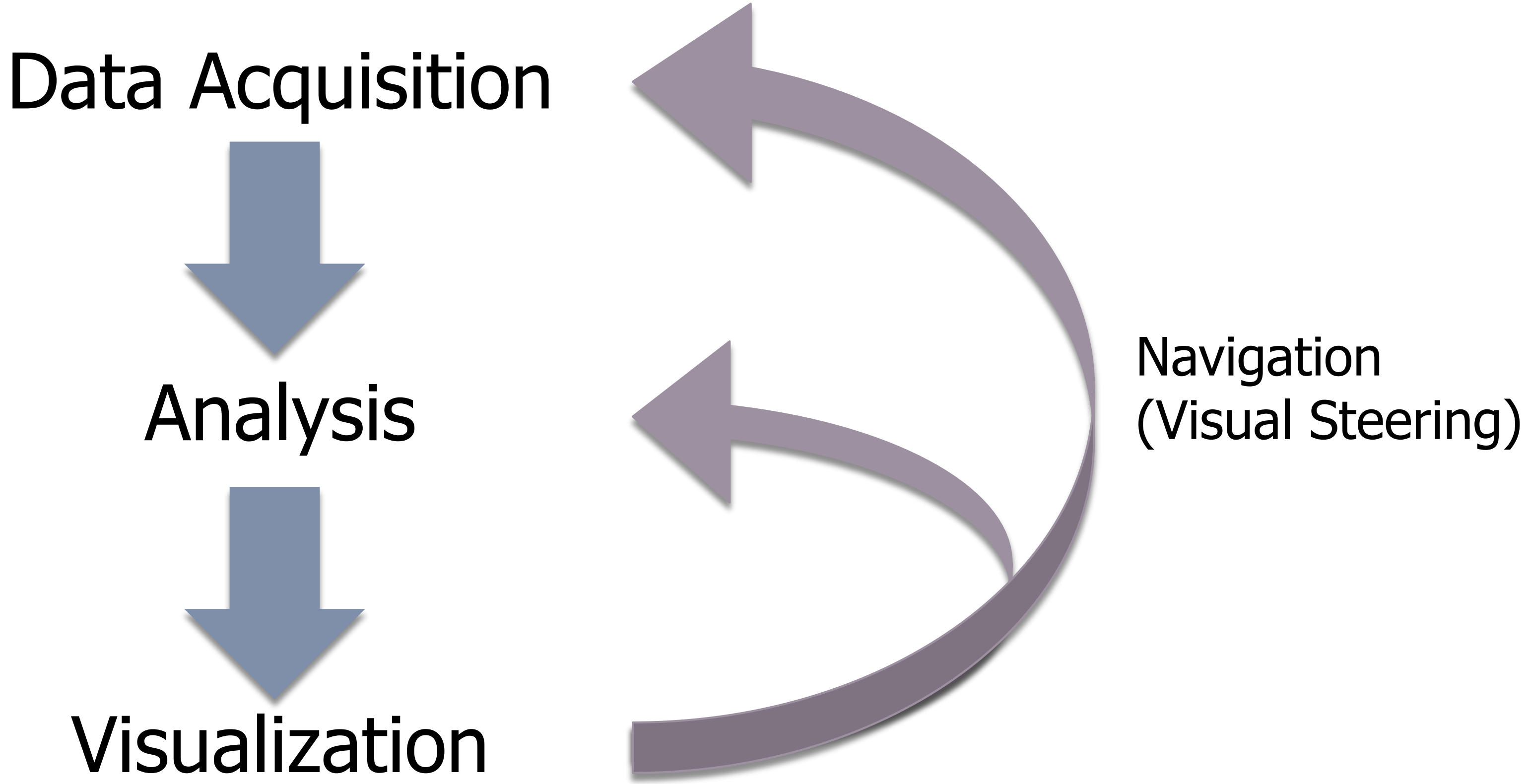
[McCormick et al.,1987]

Brainstorming

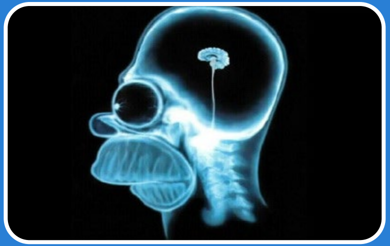
Examples of Visualizations?

Are these information visualizations?

Visualization Process



Demarcation of research areas



Scientific Visualization

- Visualization of data with a concrete physical representation (non-abstract data)

Computer Graphics

- Technical and mathematical aspects of visualization

Graphic Design

- Aesthetic graphical representation

Topics of this lecture

I. Foundations
- Info graphics
- Visual perception and cognition

II. Information visualization
- Hierarchies
- Graphs
- Multivariate data and time series

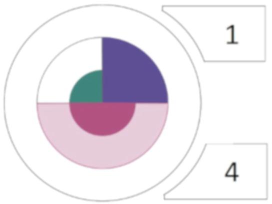
III. Related topics
- (Visual analytics)
- Human computer interaction

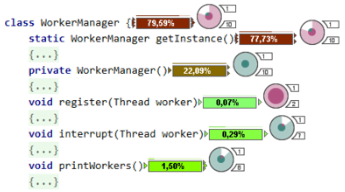
IV. Software visualization
- Code and architecture
- Behavior and evolution

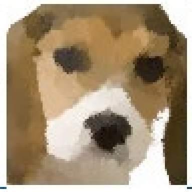
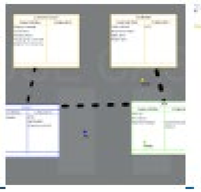
V. Conclusion
- Empirical results

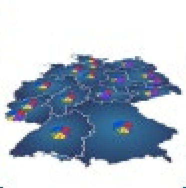
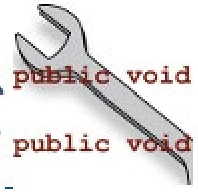
InfoVis/SoftVis-Research in Trier

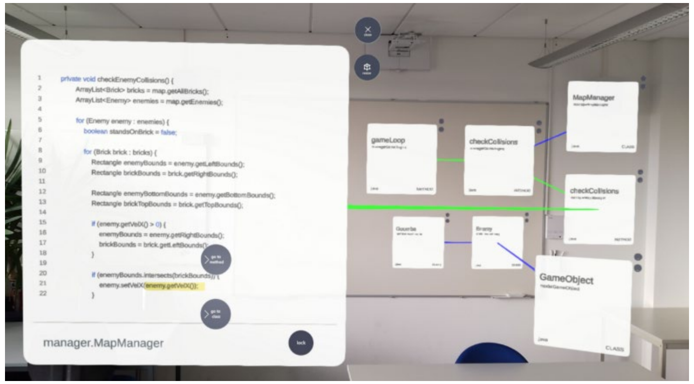
- Visualization of the evolution of software
- Visualization of (dynamic) graphs
- Comparison/merging of hierarchies and graphs
- Visualization in the IDE
- Augmented Reality for software engineering





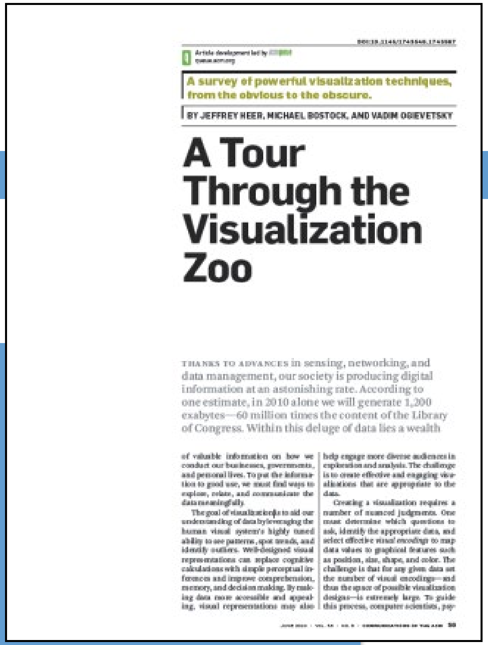






http://www.st.uni-trier.de (→ Forschung/Research)

First Exercise Sheet


A survey of powerful visualization techniques, from the obvious to the obscure.

BY JEFFREY HEER, MICHAEL BOSTOCK, AND VADIM OGIEVETSKY

A Tour Through the Visualization Zoo

THANKS TO ADVANCES in sensing, networking, and data management, our society is producing digital information at an astonishing rate. According to one estimate, in 2010 alone we will generate 1,200 exabytes—60 million times the content of the Library of Congress. Within this deluge of data lies a wealth

- **Reading**

  A Tour Through the Visualization Zoo
  Jeffrey Heer, Michael Bostock, Vadim Ogievetsky
  Communications of the ACM, Volume 53 Issue 6, June 2010

- **Task:**
  - create mind map;
  - compare visualizations in a table